Building a statistical computing textbook in the Age of AI

Structural decisions, AI-collaboration framework, and Quarto tooling across a two-volume book sequence

Ronald 'Ryy' G. Thomas

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Table of contents

1 Introduction **2**
- 1.1 Motivations . . . 3
- 1.2 Objectives . . . 3

2 What is a Quarto book? **4**

3 Prerequisites **4**

4 Installation **5**

5 Configuration: the chapter template **5**

6 The four structural decisions **7**
- 6.1 Decision 1: book boundary and overlap . . . 7
- 6.2 Decision 2: Age-of-AI framing . . . 7
- 6.3 Decision 3: sourcing chapter content . . . 7
- 6.4 Decision 4: visual differentiation . . . 8

7 Verification **8**

8 Daily Workflow **8**

9 Things to Watch Out For **9**

10 Uninstall / Rollback **10**

11 What Did We Learn? **10**
- 11.1 Lessons Learnt . . . 10
- 11.2 Limitations . . . 11
- 11.3 Opportunities for Improvement . . . 11

12 Wrapping Up 12

13 See Also 12

14 Reproducibility 13

15 Let's Connect 14

Figure 1: A shelf of two books rendered as a Quarto site, symbolising the dual-volume structure that emerged during this drafting session.

A textbook's subtitle is a commitment. 'In the Age of AI' commits the author to explaining where the human-LLM division of labour actually falls for each topic.

1 Introduction

I did not really appreciate how much structural decision-making goes into a textbook until I tried to draft two of them at once. A methods volume ('Statistical Computing in the Age of AI') and a workflow companion ('Biostatistics Practicum') were both overdue, and I wanted them to share enough scaffolding that a student could move between them without reorienting.

The turning point came when I stopped treating 'in the Age of AI' as a subtitle and started treating it as a structural obligation. Three short prose sections of 'Prompts to try' at the end of each chapter was decoration. What the subtitle required was an explicit front-loaded treatment of what the human statistician contributes, paired with an end-of-chapter verification workflow that exercises those contributions.

This post walks through the decisions that produced the two books' current state: chapter template, visual design, bibliography growth, and the specific AI-collaboration pattern. The post is a retrospective on one working session, not a finished tutorial. Readers drafting their own domain-specific textbook can probably save themselves some of the back-tracking I did.

1.1 Motivations

- Existing statistical computing textbooks are excellent but largely predate widespread LLM use. A curriculum written as if 2019 were still the current year trains students for a workplace that no longer exists.
- Graduate biostatistics students arrive with LLMs already in hand. Pretending otherwise produces a textbook that does not match their reading context.
- I wanted a dual-volume approach: a methods book (numerical algorithms, models, bootstrap, Bayesian) and a practicum (reproducibility, Git, Docker, SAS, CDISC). The two should share scaffolding but remain independently readable.
- A peer-program survey (22 US biostat MS programmes) had surfaced specific content gaps (missing data, SAS, survival, Bayesian computation) that I wanted to close without restructuring the books.
- I wanted to document the decision-making as it happened so that the next instructor building out a similar text has a worked example to follow.

1.2 Objectives

1. Scaffold two Quarto books (a methods volume and a practicum companion) with compatible chapter structure, shared citation conventions, and visually distinct rendering.
2. Establish a 'statistician's contribution' / 'collaborating with an LLM' pattern that earns the 'in the Age of AI' subtitle.
3. Ground chapter outlines in existing lecture materials, a comprehensive peer-MS-programme survey, and Jenny Bryan's STAT 545 philosophical framing.
4. Apply front-matter and code-block improvements (code annotations, first-class callouts, typographic differentiation) as a named 'polish' pass.

I am documenting the process as I learned it. If you spot mistakes or have a better approach, please let me know.

Figure 2: Workspace ambiance image: a desk with two browser tabs open side-by-side, each rendering a different Quarto book under the same file-system hierarchy.

2 What is a Quarto book?

A Quarto book is a directory of `.qmd` files (Markdown with executable R code chunks) bound together by a `_quarto.yml` configuration, rendered to a multi-format output (HTML, PDF, EPUB, Word) by the `quarto` command-line tool. Think of it as a technical-writing Makefile: you declare chapters, parts, cross-references, and output formats in YAML, and `quarto render` produces a website plus a printable PDF from the same source.

The specific convention I follow is the Posit book-family pattern (R for Data Science, Advanced R, R Packages). Each content chapter carries a three-question diagnostic quiz at the top, collapsible 'check your understanding' callouts distributed through the content, exercises and a 'further reading' block at the foot, and the quiz answers at the very end.

3 Prerequisites

This process assumed the following:

- **Operating system:** macOS 15.4, though the setup is portable to Linux and Windows.
- **R 4.4+**, RStudio optional.

- **Quarto 1.5+** (the book format changed non-trivially between 1.4 and 1.5; earlier versions will misbehave).
- **Git** and a GitHub account.
- **An LLM** used as a drafting and verification assistant. The session used Claude Code via terminal; ChatGPT or Gemini would work similarly.
- **Existing lecture materials** (speaker notes `.md`, slides `.qmd`, R code `.R`) for the methods volume; these become the source for chapter content.
- **Time required:** roughly 10 hours for the initial scaffold and framework decisions, excluding content porting.

4 Installation

No new installation was required beyond the existing Quarto and R setup. The zzcollab framework (an opinionated research-compendium scaffold) created the working directory structure automatically. The only unusual tool was a Python helper script for batch edits across 45 chapter files, written ad-hoc during the session.

```
quarto --version
# 1.9.37
R --version
# R version 4.5.3 (2026-02-28)
```

5 Configuration: the chapter template

The single most consequential decision was the chapter template. After several iterations, the structure below became the pattern for every content chapter in both books.

```
# Chapter structure (applied to every content chapter)
sections:
  - Prerequisites             # Advanced-R-style 3-question quiz
  - Learning objectives
  - Orientation
  - The statistician's contribution   # Age-of-AI pillar 1
  - Content sections                  # with Check-your-understanding callouts
  - Collaborating with an LLM on X    # Age-of-AI pillar 2
  - Exercises
  - Further reading
  - Practice test                     # only where a course test bank exists
  - Prerequisites answers             # placed at the very end
```

The two sections that earn the subtitle are 'The statistician's contribution' (front-loaded consciousness raising: the two to five decisions a statistician cannot delegate to an LLM) and 'Collaborating with an LLM on X' (end-of-chapter verification prompts, each with a Prompt / What to watch for / Verification triple).

Example of the statistician's-contribution section for the bootstrap chapter:

```
## The statistician's contribution {#sec-bootstrap-human}

Before handing a dataset to a large language model with the
instruction to 'bootstrap the standard error', four decisions
require the statistician's judgment:

1. Is the bootstrap appropriate for this statistic at all?
2. What is the dependence structure of the data?
3. Which confidence-interval method matches the statistic's
   behaviour?
4. Is the bootstrap distribution itself plausible?
```

Each decision gets a paragraph of explanation, with the failure modes named explicitly (extrema, time-series data, skewed bootstrap distributions) and the chapter that follows providing the vocabulary to make each decision.

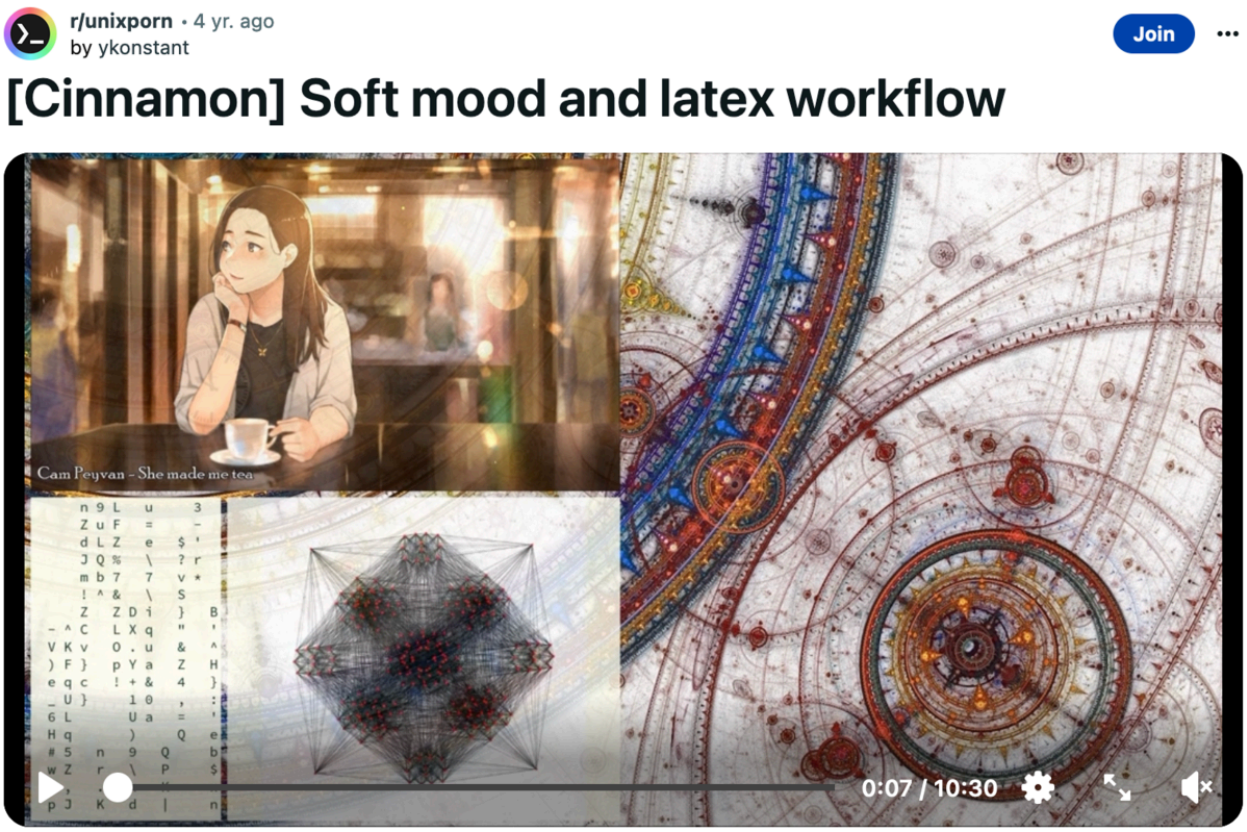

Figure 3: Terminal showing the book's sidebar in the browser, with the two books' shared chapter structure visible as identical tree layouts under different part-labels.

6 The four structural decisions

The session's work sorted into four structural decisions, each with downstream consequences across 45 chapters.

6.1 Decision 1: book boundary and overlap

The methods volume covers statistical-theory content an MS programme teaches in a typical one-quarter course: programming in R, numerical linear algebra, optimisation, simulation, bootstrap, linear models, GLM, mixed models, survival, Bayesian. The practicum covers the workflow that surrounds the methods: reproducibility infrastructure, Git, Docker, rrtools, renv, Quarto, tidyverse wrangling, CDISC data standards, SAS, and two full case studies.

The boundary is enforced by discipline in the table of contents. Where a chapter in one volume touches a topic covered in the other, the prose uses a textual pointer ('see the Missing Data chapter of the companion *Biostatistics Practicum* volume') rather than a Quarto cross-reference, because `@sec-*` anchors do not resolve across distinct books.

6.2 Decision 2: Age-of-AI framing

The original chapter template ended with a single 'With AI assistance' callout containing three prompts. Reviewing the material as a whole showed that three decorative prompts do not earn the 'in the Age of AI' subtitle; they behave as a sidebar rather than as part of the intellectual framework.

The replacement is two structured sections per chapter: a front-loaded 'The statistician's contribution' (what the LLM cannot do on the reader's behalf) and an end-of-chapter 'Collaborating with an LLM on X' (what to verify when the LLM produces output). The preface and introduction were rewritten to name the four-category framework (reliable, unreliable, cannot-due-to-missing-context, cannot-due-to-accountability) that organises the book's position.

6.3 Decision 3: sourcing chapter content

Each chapter's source material is distributed across three file types: a `speaker_notes_expanded.md` that carries the prose narrative, a `lecture*_slides_*.qmd` that carries both inline code blocks and 'dynamic' `.fragment .question` / `.answer` pairs, and a `lecture*_R_code.R` with supplementary code examples. Early chapter ports missed the inline slide code and the fragment pairs entirely because the porting workflow only read the speaker notes and the R file.

The revised workflow reads all three sources, integrates inline code from the slides rather than from the stand-alone R file (the two sometimes disagree), and converts each slide's dynamic Q&A fragment into a collapsible 'Check your understanding' callout placed after the relevant content section.

6.4 Decision 4: visual differentiation

Opening the two books in adjacent browser tabs, they were indistinguishable at a glance. Both used the same primary colour, the same body face, the same monospace face, the same SCSS rules. The fix was a deliberate typographic split: the methods volume renders body text in Source Serif 4 (fallbacks Charter, Georgia), the practicum renders body text in Source Sans 3 (fallbacks Inter, Helvetica Neue). Monospace stays JetBrains Mono in both. The SCSS change was ten lines per book.

Complementary configuration changes in `_quarto.yml`:

```yaml
format:
  html:
    code-annotations: hover
    code-copy: hover
    code-overflow: wrap
    code-tools:
      source: repo
      toggle: false
    code-block-bg: true
    code-block-border-left: '#1f4e79'
    highlight-style: arrow
```

Each of these settings is a named Quarto feature. `code-annotations: hover` enables numbered-disc annotations inline in code. `code-tools` puts a 'view source' chevron on each page that links to the source `.qmd` in the book's GitHub repository. `code-block-border-left` replaces a hand-rolled SCSS rule with Quarto-native styling synchronised with dark-mode theming. `highlight-style: arrow` is the accessibility-tuned palette and renders legibly in both HTML and PDF.

7 Verification

After every structural change, I re-rendered both books and checked the warnings:

```bash
cd path/to/methods/textbook
quarto render --to html 2>&1 | grep -iE 'error|warn'

cd path/to/practicum
quarto render --to html 2>&1 | grep -iE 'error|warn'
```

Clean output (empty `grep` result) is the signal to continue. Citeproc warnings ('citation X not found') always flagged a bibliography gap and were fixed immediately; bib entries were added incrementally as content ported in.

8 Daily Workflow

The tools that came up most often:

Command	Action
`quarto render --to html`	Full book render (HTML only, fast)
`quarto render path/to/chapter.qmd`	Single chapter render for rapid iteration
`quarto preview`	Live-reload preview in browser
`quarto check knitr`	Confirm the R engine is configured
`grep -cE '^@' references.bib`	Count bibliography entries (growth log)
`ls [0-9][0-9]-*.qmd` + `wc -l`	Confirm expected chapter count

The bulk batch-edit pattern (rename a heading across 45 chapters, insert a section block after a given anchor) was handled by short ad-hoc Python scripts using `pathlib` and a content-map dictionary. Shell `sed` worked for single-line substitutions but became unmanageable for multi-line insertions.

Figure 4: A workflow scene showing a simple diagram of the two books' structure: seven parts in the methods volume, seven parts in the practicum, each containing the same chapter template.

9 Things to Watch Out For

1. **Quarto's auto-numbering silently breaks if the introduction chapter is not unnumbered.** `00-intro.qmd` without `.unnumbered` gets counted as Chapter 1, shifting every

subsequent chapter's rendered number up by one. Filename prefix says `09-bootstrap.qmd`, browser says 'Chapter 10'. Symptom: every cross-reference in the text reads one number off. Fix: `# Introduction {#sec-intro .unnumbered}`.

2. **Subsection numeric prefixes collide with Quarto's auto-numbering.** Writing `### 1. Correlation and the Fisher-z transformation` renders as '9.12.1 1. Correlation…'. Fix: drop the prefix, let Quarto number.

3. **Missing bibliography entries warn but do not fail.** `[WARNING] Citeproc: citation mcelreath2020rethinking not found` is easy to miss in a full-render log. Pipe the render through `grep -iE 'error|warn'` every time.

4. **Cross-book references do not resolve.** `@sec-cdisc` in the methods volume cannot find the anchor if that anchor lives in the practicum volume. Fix: use prose pointers ('see the CDISC chapter of the companion volume').

5. **The `Quiz answers` heading label mismatches the `Prerequisites` quiz label.** Advanced R uses 'Quiz' / 'Quiz answers'; my chapters used 'Prerequisites' / 'Quiz answers', which readers found confusing. Fix: rename the bottom section 'Prerequisites answers' and move it to the very end of the chapter (after the Practice test, where one exists).

6. **Speaker notes, slide deck, and R code file each carry different content for the same lecture.** Slide `.qmd` files are the primary source for inline code; the stand-alone `.R` file often omits blocks that only appear inline in the deck. Read all three.

7. **Dynamic `.fragment .question` / `.answer` blocks in slides are easy to miss.** These are the interactive 'check your understanding' moments in a live lecture. In book form they become collapsible callouts. Grep for `fragment .question` across the lecture directories to find them.

10 Uninstall / Rollback

Nothing installed that needed rollback. All work is in Git; `git reset --hard <commit>` undoes any session.

11 What Did We Learn?

11.1 Lessons Learnt

Conceptual Understanding:

- A subtitle is a structural commitment, not a tagline. 'In the Age of AI' obliged the book to explain where the human-LLM division of labour falls for each topic. Three prose prompts at the end of a chapter did not discharge the commitment.
- Two books are easier to draft than one, because the scope limits become explicit. Each volume has a boundary, and content that straddles the boundary gets a textual pointer rather than a cross-book link.

- Peer programs converge on a fairly consistent MS curriculum. A 22-programme survey found 17/22 require SAS, 14/22 require missing data, 13/22 require Bayesian computation. Content gaps close faster when the reference distribution is visible.
- The Posit book family (r4ds, r-pkgs, Advanced R, ggplot2) is an excellent structural template, but its conventions do not by themselves earn an 'Age of AI' subtitle. That work is additive.

Technical Skills:

- Quarto's code-annotation feature is under-used. It is the single most distinctive affordance for a computational textbook and was off by default in both books.
- Visual differentiation costs almost nothing: a ten-line SCSS change gives each of two sibling books a distinct body face.
- Collapsible callouts (`.callout-tip collapse='true'`) are the ideal home for slide-style 'check your understanding' fragments in a book context. The reader sees the question, thinks, then clicks to reveal the answer.
- Bibliography growth should track content porting. Retrofitting a `references.bib` at the end is painful; adding entries as chapters fill in is trivial.

Gotchas and Pitfalls:

- The `.unnumbered` class on introduction chapters is load-bearing. Without it, every subsequent chapter renders with a shifted number.
- Subsection numeric prefixes duplicate Quarto's auto-numbering ('9.12.1 1.'). Let Quarto do the numbering.
- Tidyverse-style `@sec-*` cross-references are invisible in a rendered TOC. They resolve correctly but only in the body text.
- Cross-book `@sec-*` anchors do not resolve. Use prose pointers.

11.2 Limitations

- Single author, not peer-reviewed. Content decisions were informed by an LLM assistant but not by a disciplinary editorial board.
- Most chapters are still scaffolds. The skeleton (Prerequisites, Learning Objectives, Exercises, Further Reading, Quiz answers) is complete, but the interior content sections are TODO placeholders in all but a handful of chapters.
- The bootstrap chapter was ported end-to-end as a reference implementation. The remaining 19 methods chapters and most practicum chapters await a similar pass.
- Practice tests are only present in chapters where the course's test bank contains matching material. New chapters (Bayesian, survival) have no practice test by design.
- No automated progress tracking. A simple markdown table mapping chapter to 'scaffold / ported / polished' status would make collaboration easier.

11.3 Opportunities for Improvement

1. Port the remaining 19 content chapters' speaker notes, slide code, and dynamic Q&A fragments into chapter bodies following the bootstrap chapter's workflow.

2. Implement the four named callouts (`Note` / `Verification` / `Pitfall` / `LLM prompt`) consistently, migrating existing prose callouts as each chapter is touched.
3. Migrate the 'Check your understanding' Q&A fragments from untitled `callout-tip collapse='true'` to the named `callout-tip title='Check'` form so they pattern-match across chapters.
4. Add a colophon page to each book documenting the build environment, theme, and font choices.
5. Source hero and portrait images (Efron, Bates, Bryan, Wickham, Marwick) and drop them at `images/cover.png` and `images/portraits/*.jpg`.
6. Push the two book repositories to GitHub (`rgt47/scai`, `rgt47/practicum`) so the `code-tools` 'view source' chevrons resolve.
7. Configure Netlify deploys at `rgtlab.org/scai` and `rgtlab.org/practicum` via a path-based proxy from the main site.

12 Wrapping Up

A textbook's subtitle is a commitment. 'Statistical Computing in the Age of AI' obliged the book to explain where the human-LLM division of labour falls for each topic, and the explanation had to be structural rather than decorative. The current template delivers that through two sections per chapter (statistician's contribution plus collaborating with an LLM) backed by a four-category framework in the introduction.

The secondary lesson was that a dual-volume approach clarifies scope. One book tries to be everything to everyone; two books can afford to be boundary-conscious, each pointing to the other where their topics connect. The methods volume covers what modelling looks like; the practicum covers everything that surrounds modelling. A student who reads both comes out with a complete picture and has an easier time than the single-volume reader because each book's scope is visibly narrower.

Main takeaways:

- A subtitle is a structural obligation; earn it with sections, not sidebars.
- Two books with shared scaffolding and distinct visual identity are easier to maintain than one oversized volume.
- Quarto's code-annotation, code-tools, and first-class callout features should be configured on day one, not bolted on during a polish pass.
- Read the slides, the speaker notes, and the R scripts separately when porting lecture content. They do not fully overlap.

13 See Also

Related posts:

- Setup Post Template (AWS CLI): the post template this one follows.
- Template Post for Data Analysis: the data-analysis sibling template.

Key resources:

- Quarto Books documentation
- Advanced R by Hadley Wickham: the chapter-structure reference.
- R for Data Science, 2nd ed.: tidyverse applied reference.
- STAT 545 by Jenny Bryan and Derek Stephens: 'everything in data analysis except modelling' framing.
- zzcollab framework: the five-pillar research compendium scaffold used as this post's working directory.

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14 Reproducibility

Tested configuration:

Component	Version
Operating system	macOS 15.4
Quarto	1.9.37
R	4.5.3
knitr	1.51
rmarkdown	2.31
Shell	zsh 5.9
Last verified	2026-04-24

Book repositories:

- Methods volume: `~/prj/tch/methods-textbook/textbook/`
- Practicum volume: `~/prj/tch/biostatistics-practicum/`

To reproduce the two-book scaffold:

```
# Methods volume
mkdir path/to/methods-book && cd path/to/methods-book
quarto create project book
# Edit _quarto.yml with the chapter-template sections.

# Practicum volume (same)
mkdir path/to/practicum-book && cd path/to/practicum-book
quarto create project book
# Edit _quarto.yml with the chapter-template sections.
```

The chapter-template structure (Prerequisites, Learning Objectives, Orientation, The statistician's contribution, content, Check-your-understanding callouts, Collaborating with an LLM, Exercises, Further reading, Practice test, Prerequisites answers) is a narrative convention, not a Quarto feature; applying it is manual.

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15 Let's Connect

Have questions, suggestions, or spot an error? Let me know.

- **GitHub:** rgt47
- **Twitter/X:** @rgt47
- **LinkedIn:** Ronald Glenn Thomas
- **Email:** Contact form

I would enjoy hearing from you if:

- You are drafting a similar domain-specific textbook and want to compare structural decisions.
- You have a better approach to the human-LLM division of labour framework.
- You teach a biostatistics MS programme with different curriculum gaps than the ones this process surfaced.
- You just want to say hello and connect.